Exp 5. Spam Email Detection: Build a spam email filter using text classification algorithms and perform comparative analysis. (Naïve Bayes, Logistic Regression, Support Vector Machine, Decision Tree, Random Forest)

Note: Use Evaluation Metrics (Accuracy, Precision, Recall, F1-Score for binary classification), (ROC-AUC, Accuracy, Precision, Recall, Weighted / Macro/ Micro Average F1-Score for Multiclassification task) and Confusion Matrix.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score,precision_score,recall_score,f1_score,confusion_matrix,classification_report,roc_auc_score
import pickle
```

```
df=pd.read_csv('/content/drive/MyDrive/spam_email_500_samples.csv')
df.head()
```

	label	text
0	spam	Free entry in 2 million dollar contest. Click ...
1	spam	Act now! This offer is valid for today only.
2	spam	Get cheap medicines online without prescriptio...
3	ham	Lunch at 1 PM works for me.
4	ham	Thank you for your assistance.

Next steps: Generate code with df New interactive sheet

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   label   500 non-null    object
 1   text    500 non-null    object
dtypes: object(2)
memory usage: 7.9+ KB
```

	label	text
count	500	500
unique	2	20
top	spam	Get cheap medicines online without prescriptio...
freq	250	40

```
df['label']=df['label'].map({'ham':0,'spam':1})
df.head()
```

	label	text
0	1	Free entry in 2 million dollar contest. Click ...
1	1	Act now! This offer is valid for today only.
2	1	Get cheap medicines online without prescriptio...
3	0	Lunch at 1 PM works for me.
4	0	Thank you for your assistance.

Next steps: Generate code with df New interactive sheet

```
X=df['text']
y=df['label']
```

```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=0)
```

```
cv=CountVectorizer(stop_words='english')
X_train=cv.fit_transform(X_train)
X_test=cv.transform(X_test)
print("Training matrix shape:",X_train.shape)
print("Testing matrix shape:",X_test.shape)
```

```
Training matrix shape: (400, 83)
Testing matrix shape: (100, 83)
```

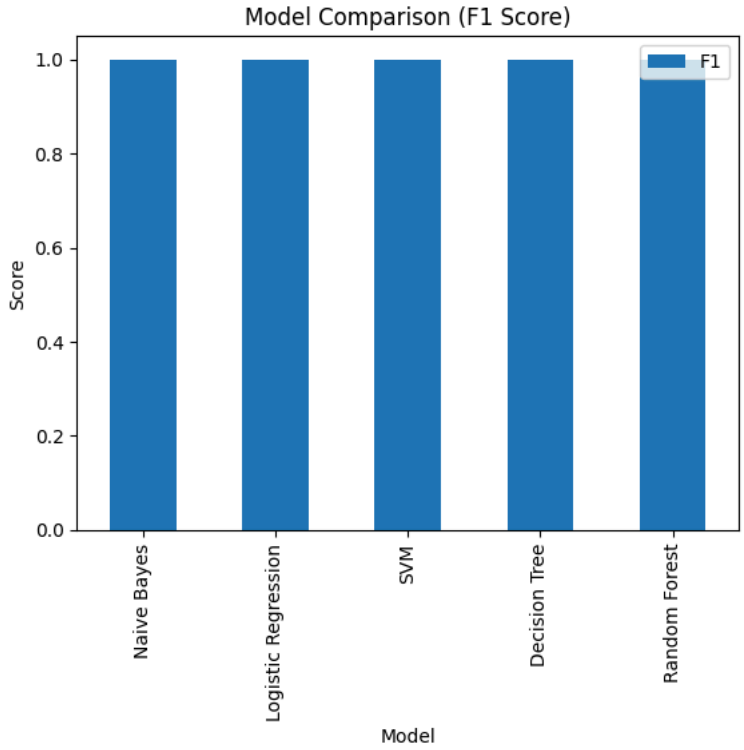
```
models={"Naive Bayes":MultinomialNB(),
        "Logistic Regression":LogisticRegression(max_iter=100),
        "SVM":SVC(probability=True),
        "Decision Tree":DecisionTreeClassifier(),
        "Random Forest":RandomForestClassifier()
        }
```

```
results=[]
for name,model in models.items():
    model.fit(X_train,y_train)
    pred=model.predict(X_test)
    acc=accuracy_score(y_test,pred)
    pre=precision_score(y_test,pred)
    rec=recall_score(y_test,pred)
    f1=f1_score(y_test,pred)
    results.append([name,acc,pre,rec,f1])
results_df=pd.DataFrame(results,columns=["Model","Accuracy","Precision","Recall","F1"])
results_df
```

	Model	Accuracy	Precision	Recall	F1
0	Naive Bayes	1.0	1.0	1.0	1.0
1	Logistic Regression	1.0	1.0	1.0	1.0
2	SVM	1.0	1.0	1.0	1.0
3	Decision Tree	1.0	1.0	1.0	1.0
4	Random Forest	1.0	1.0	1.0	1.0

Next steps: Generate code with results_df New interactive sheet

```
results_df.plot(x="Model",y="F1",kind="bar")
plt.title("Model Comparison (F1 Score)")
plt.ylabel("Score")
plt.show()
```



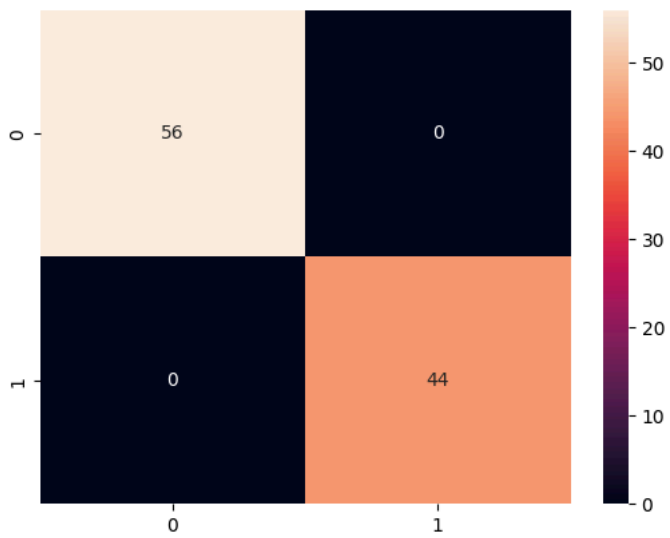

```
best_name=results_df.sort_values("F1",ascending=False).iloc[0,0]
best_model=models[best_name]
print("Best Model:",best_name)
```

```
Best Model: Naive Bayes
```

```
with open("spam_email_classifier.pkl","wb") as f:
    pickle.dump(best_model,f)
```

```
pred=best_model.predict(X_test)
cm=confusion_matrix(y_test,pred)
sns.heatmap(cm,annot=True,fmt="d")
```

```
<Axes: >
```




```
probs=best_model.predict_proba(X_test)[:,1]
print("ROC AUC:",roc_auc_score(y_test,probs))
```

```
ROC AUC: 1.0
```

```
print(classification_report(y_test,pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        56
           1       1.00      1.00      1.00        44

    accuracy                           1.00       100
   macro avg       1.00      1.00      1.00       100
weighted avg       1.00      1.00      1.00       100
```